

Class 10 Science — Tough Practice Set

100 Questions | Level: Tough

StudyShelf — studysshelf practice worksheet. Answer key provided at the end.

- Q1.** Explain why carbon cannot form $C4^{+}$ or $C4^{-}$ ions easily and how it satisfies its valency instead.
- Q2.** Explain the difference between nuclear fission and nuclear fusion, giving one example of each.
- Q3.** An object is placed at a distance of 40 cm in front of a concave mirror of focal length 30 cm. Find the image distance using the mirror formula.
- Q4.** Find the heat produced when a current of 5 A flows through a resistor of $20\ \Omega$ for 300 seconds.
- Q5.** Find the heat produced when a current of 1 A flows through a resistor of $15\ \Omega$ for 300 seconds.
- Q6.** Find the heat produced when a current of 5 A flows through a resistor of $20\ \Omega$ for 30 seconds.
- Q7.** An object is placed at a distance of 35 cm in front of a concave mirror of focal length 20 cm. Find the image distance using the mirror formula.
- Q8.** Describe the process of nutrition in Amoeba and relate it to the general process of digestion.
- Q9.** Explain how variation is important for the survival of a species, using Darwin's theory as reference.
- Q10.** Find the heat produced when a current of 2 A flows through a resistor of $5\ \Omega$ for 120 seconds.
- Q11.** Find the heat produced when a current of 2 A flows through a resistor of $15\ \Omega$ for 120 seconds.
- Q12.** Explain the mechanism of double circulation in humans and why it is more efficient than single circulation.
- Q13.** Find the heat produced when a current of 5 A flows through a resistor of $5\ \Omega$ for 30 seconds.
- Q14.** With the help of a suitable example, explain why a chemical equation must be balanced.
- Q15.** An object is placed at a distance of 45 cm in front of a concave mirror of focal length 25 cm. Find the image distance using the mirror formula.
- Q16.** Find the heat produced when a current of 5 A flows through a resistor of $20\ \Omega$ for 120 seconds.
- Q17.** An object is placed at a distance of 15 cm in front of a concave mirror of focal length 10 cm. Find the image distance using the mirror formula.
- Q18.** Find the heat produced when a current of 1 A flows through a resistor of $10\ \Omega$ for 30 seconds.
- Q19.** An object is placed at a distance of 30 cm in front of a concave mirror of focal length 20 cm. Find the image distance using the mirror formula.
- Q20.** An object is placed at a distance of 50 cm in front of a concave mirror of focal length 30 cm. Find the image distance using the mirror formula.
- Q21.** Discuss how the structure of alveoli is suited to their function of gas exchange in the lungs.
- Q22.** Explain the significance of the ozone layer and describe one major cause of its depletion.

Q23. Find the heat produced when a current of 4 A flows through a resistor of $15\ \Omega$ for 300 seconds.

Q24. An object is placed at a distance of 40 cm in front of a concave mirror of focal length 20 cm. Find the image distance using the mirror formula.

Q25. An object is placed at a distance of 30 cm in front of a concave mirror of focal length 25 cm. Find the image distance using the mirror formula.

Q26. Find the heat produced when a current of 5 A flows through a resistor of $15\ \Omega$ for 60 seconds.

Q27. Find the heat produced when a current of 4 A flows through a resistor of $20\ \Omega$ for 120 seconds.

Q28. Explain why the tip of a root always grows towards gravity while the shoot grows away from it.

Q29. An object is placed at a distance of 45 cm in front of a concave mirror of focal length 30 cm. Find the image distance using the mirror formula.

Q30. Explain how the human eye is able to focus on both near and distant objects, and what is this ability called.

Q31. A metal 'M' reacts with dilute HCl to produce brisk effervescence, but does not react with dilute NaOH. Identify the possible nature of the metal and justify.

Q32. Find the heat produced when a current of 1 A flows through a resistor of $20\ \Omega$ for 120 seconds.

Q33. Find the heat produced when a current of 5 A flows through a resistor of $10\ \Omega$ for 120 seconds.

Q34. Find the heat produced when a current of 2 A flows through a resistor of $5\ \Omega$ for 60 seconds.

Q35. An object is placed at a distance of 40 cm in front of a concave mirror of focal length 25 cm. Find the image distance using the mirror formula.

Q36. Find the heat produced when a current of 3 A flows through a resistor of $10\ \Omega$ for 30 seconds.

Q37. Find the heat produced when a current of 2 A flows through a resistor of $20\ \Omega$ for 30 seconds.

Q38. Find the heat produced when a current of 1 A flows through a resistor of $5\ \Omega$ for 60 seconds.

Q39. An object is placed at a distance of 20 cm in front of a concave mirror of focal length 10 cm. Find the image distance using the mirror formula.

Q40. Describe the working of an electric generator and state the principle it is based on.

Q41. An object is placed at a distance of 25 cm in front of a concave mirror of focal length 15 cm. Find the image distance using the mirror formula.

Q42. An object is placed at a distance of 25 cm in front of a concave mirror of focal length 10 cm. Find the image distance using the mirror formula.

Q43. An object is placed at a distance of 30 cm in front of a concave mirror of focal length 15 cm. Find the image distance using the mirror formula.

Q44. Find the heat produced when a current of 4 A flows through a resistor of $10\ \Omega$ for 300 seconds.

Q45. An object is placed at a distance of 20 cm in front of a concave mirror of focal length 15 cm. Find the image distance using the mirror formula.

- Q46.** An object is placed at a distance of 30 cm in front of a concave mirror of focal length 10 cm. Find the image distance using the mirror formula.
- Q47.** An object is placed at a distance of 35 cm in front of a concave mirror of focal length 25 cm. Find the image distance using the mirror formula.
- Q48.** Find the heat produced when a current of 2 A flows through a resistor of $20\ \Omega$ for 120 seconds.
- Q49.** An object is placed at a distance of 35 cm in front of a concave mirror of focal length 15 cm. Find the image distance using the mirror formula.
- Q50.** Find the heat produced when a current of 1 A flows through a resistor of $15\ \Omega$ for 60 seconds.
- Q51.** An object is placed at a distance of 25 cm in front of a concave mirror of focal length 20 cm. Find the image distance using the mirror formula.
- Q52.** Find the heat produced when a current of 3 A flows through a resistor of $10\ \Omega$ for 120 seconds.
- Q53.** Find the heat produced when a current of 5 A flows through a resistor of $15\ \Omega$ for 120 seconds.
- Q54.** Find the heat produced when a current of 1 A flows through a resistor of $15\ \Omega$ for 30 seconds.
- Q55.** Find the heat produced when a current of 4 A flows through a resistor of $15\ \Omega$ for 60 seconds.
- Q56.** Find the heat produced when a current of 3 A flows through a resistor of $15\ \Omega$ for 120 seconds.
- Q57.** Find the heat produced when a current of 3 A flows through a resistor of $5\ \Omega$ for 120 seconds.
- Q58.** Find the heat produced when a current of 2 A flows through a resistor of $10\ \Omega$ for 300 seconds.
- Q59.** Find the heat produced when a current of 5 A flows through a resistor of $10\ \Omega$ for 300 seconds.
- Q60.** Find the heat produced when a current of 2 A flows through a resistor of $5\ \Omega$ for 30 seconds.
- Q61.** An object is placed at a distance of 35 cm in front of a concave mirror of focal length 30 cm. Find the image distance using the mirror formula.
- Q62.** Find the heat produced when a current of 1 A flows through a resistor of $20\ \Omega$ for 30 seconds.
- Q63.** Find the heat produced when a current of 3 A flows through a resistor of $15\ \Omega$ for 60 seconds.
- Q64.** Find the heat produced when a current of 3 A flows through a resistor of $20\ \Omega$ for 120 seconds.
- Q65.** Find the heat produced when a current of 2 A flows through a resistor of $10\ \Omega$ for 30 seconds.
- Q66.** Find the heat produced when a current of 4 A flows through a resistor of $10\ \Omega$ for 30 seconds.
- Q67.** Find the heat produced when a current of 3 A flows through a resistor of $15\ \Omega$ for 300 seconds.
- Q68.** Find the heat produced when a current of 5 A flows through a resistor of $15\ \Omega$ for 30 seconds.
- Q69.** Find the heat produced when a current of 1 A flows through a resistor of $15\ \Omega$ for 120 seconds.
- Q70.** Find the heat produced when a current of 3 A flows through a resistor of $10\ \Omega$ for 60 seconds.
- Q71.** Find the heat produced when a current of 5 A flows through a resistor of $5\ \Omega$ for 300 seconds.

- Q72.** Find the heat produced when a current of 1 A flows through a resistor of 5 Ω for 300 seconds.
- Q73.** Find the heat produced when a current of 2 A flows through a resistor of 20 Ω for 300 seconds.
- Q74.** Find the heat produced when a current of 2 A flows through a resistor of 15 Ω for 30 seconds.
- Q75.** Find the heat produced when a current of 5 A flows through a resistor of 5 Ω for 120 seconds.
- Q76.** Find the heat produced when a current of 4 A flows through a resistor of 10 Ω for 120 seconds.
- Q77.** Find the heat produced when a current of 4 A flows through a resistor of 5 Ω for 300 seconds.
- Q78.** Find the heat produced when a current of 1 A flows through a resistor of 10 Ω for 300 seconds.
- Q79.** Find the heat produced when a current of 3 A flows through a resistor of 15 Ω for 30 seconds.
- Q80.** Find the heat produced when a current of 4 A flows through a resistor of 5 Ω for 60 seconds.
- Q81.** Find the heat produced when a current of 3 A flows through a resistor of 5 Ω for 60 seconds.
- Q82.** Find the heat produced when a current of 2 A flows through a resistor of 10 Ω for 60 seconds.
- Q83.** Find the heat produced when a current of 4 A flows through a resistor of 20 Ω for 30 seconds.
- Q84.** Find the heat produced when a current of 5 A flows through a resistor of 10 Ω for 30 seconds.
- Q85.** Find the heat produced when a current of 3 A flows through a resistor of 5 Ω for 30 seconds.
- Q86.** Find the heat produced when a current of 2 A flows through a resistor of 15 Ω for 60 seconds.
- Q87.** Find the heat produced when a current of 1 A flows through a resistor of 10 Ω for 120 seconds.
- Q88.** Find the heat produced when a current of 1 A flows through a resistor of 10 Ω for 60 seconds.
- Q89.** Find the heat produced when a current of 4 A flows through a resistor of 15 Ω for 120 seconds.
- Q90.** Find the heat produced when a current of 4 A flows through a resistor of 5 Ω for 120 seconds.
- Q91.** Find the heat produced when a current of 2 A flows through a resistor of 10 Ω for 120 seconds.
- Q92.** Find the heat produced when a current of 3 A flows through a resistor of 20 Ω for 300 seconds.
- Q93.** Find the heat produced when a current of 5 A flows through a resistor of 5 Ω for 60 seconds.
- Q94.** Find the heat produced when a current of 1 A flows through a resistor of 5 Ω for 30 seconds.
- Q95.** Find the heat produced when a current of 4 A flows through a resistor of 20 Ω for 60 seconds.
- Q96.** Find the heat produced when a current of 4 A flows through a resistor of 5 Ω for 30 seconds.
- Q97.** Find the heat produced when a current of 5 A flows through a resistor of 20 Ω for 60 seconds.
- Q98.** Find the heat produced when a current of 4 A flows through a resistor of 20 Ω for 300 seconds.
- Q99.** Find the heat produced when a current of 3 A flows through a resistor of 20 Ω for 30 seconds.
- Q100.** Find the heat produced when a current of 3 A flows through a resistor of 10 Ω for 300 seconds.

Answer Key

1. Removing 4 electrons (C^{4+}) would require very high energy, and gaining 4 electrons (C^{4-}) is difficult to hold onto due to the small size of carbon's nucleus attracting them weakly; instead, carbon satisfies its valency of 4 by sharing electrons, forming covalent bonds.

2. Nuclear fission is the splitting of a heavy nucleus into smaller nuclei with release of energy, e.g., splitting of Uranium-235 in a nuclear reactor. Nuclear fusion is the combining of light nuclei to form a heavier nucleus with release of energy, e.g., fusion of hydrogen nuclei in the Sun.

3. $v \approx 17.14 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

4. $H = I^2 R t = 150000 \text{ J}$

5. $H = I^2 R t = 4500 \text{ J}$

6. $H = I^2 R t = 15000 \text{ J}$

7. $v \approx 12.73 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

8. Amoeba engulfs food using pseudopodia to form a food vacuole (phagocytosis), where enzymes digest the food; the digested nutrients are absorbed into the cytoplasm and undigested matter is expelled — similar in principle to ingestion, digestion, absorption and egestion in higher animals.

9. Variations arise during reproduction due to errors in DNA copying; individuals with variations suited to the environment survive and reproduce better (natural selection), which over generations leads to evolution and helps species adapt to changing environments, increasing survival chances.

10. $H = I^2 R t = 2400 \text{ J}$

11. $H = I^2 R t = 7200 \text{ J}$

12. In double circulation, blood passes through the heart twice in one cycle — once for pulmonary circulation (heart to lungs and back) and once for systemic circulation (heart to body and back). This keeps oxygenated and deoxygenated blood separate, allowing more efficient oxygen supply to tissues than single circulation.

13. $H = I^2 R t = 3750 \text{ J}$

14. A chemical equation must be balanced to satisfy the Law of Conservation of Mass — the number of atoms of each element must be equal on both sides. For example, unbalanced $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$ must be balanced to $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ so atoms of Fe, H and O match on both sides.

15. $v \approx 16.07 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

16. $H = I^2 R t = 60000 \text{ J}$

17. $v \approx 6.0 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

18. $H = I^2 R t = 300 \text{ J}$

19. $v \approx 12.0 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

20. $v \approx 18.75 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

21. Alveoli are thin-walled (single layer of cells) and richly supplied with blood capillaries, and are present in large numbers, providing a very large surface area — these features together enable efficient diffusion of oxygen into blood and carbon dioxide out of blood.

22. The ozone layer in the stratosphere absorbs most of the Sun's harmful ultraviolet (UV) radiation, protecting living organisms from UV-related damage such as skin cancer and mutations. A major cause of its depletion is the release of chlorofluorocarbons (CFCs) from refrigerants and aerosols, which break down ozone molecules.

23. $H = I^2 R t = 72000 \text{ J}$

24. $v \approx 13.33 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

25. $v \approx 13.64 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

26. $H = I^2 R t = 22500 \text{ J}$

27. $H = I^2 R t = 38400 \text{ J}$

28. This is due to hormonal response to gravity called geotropism — auxin accumulates more on the lower side of the root, inhibiting its growth there so the root curves downward; in shoots, higher auxin concentration promotes growth on the lower side, causing the shoot to bend upward, away from gravity.

29. $v \approx 18.0 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

30. The ciliary muscles change the curvature (and hence focal length) of the eye lens — this ability is called accommodation. For near objects, the lens becomes more curved (thicker); for distant objects, it becomes flatter.

31. The metal is likely a reactive metal such as zinc, magnesium, or aluminium type behaviour towards acid, but not amphoteric enough to react with NaOH unless it's specifically aluminium/zinc (which are amphoteric and DO react with NaOH) — so more specifically, metal M reacting only with acid and not base suggests a metal like magnesium or iron, which reacts with dilute acids releasing H₂ gas but does not react with a base.

32. $H = I^2 R t = 2400 \text{ J}$

33. $H = I^2 R t = 30000 \text{ J}$

34. $H = I^2 R t = 1200 \text{ J}$

35. $v \approx 15.38 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

36. $H = I^2 R t = 2700 \text{ J}$

37. $H = I^2 R t = 2400 \text{ J}$

38. $H = I^2 R t = 300 \text{ J}$

39. $v \approx 6.67 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

40. An electric generator is based on the principle of electromagnetic induction — when a coil is rotated in a magnetic field, the change in magnetic flux through the coil induces an alternating EMF, which drives a current in an external circuit.

41. $v \approx 9.38 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

42. $v \approx 7.14 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

43. $v \approx 10.0 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

44. $H = I^2 R t = 48000 \text{ J}$

45. $v \approx 8.57 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

46. $v \approx 7.5 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

47. $v \approx 14.58 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

48. $H = I^2 R t = 9600 \text{ J}$

49. $v \approx 10.5 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

50. $H = I^2 R t = 900 \text{ J}$

51. $v \approx 11.11 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

52. $H = I^2 R t = 10800 \text{ J}$

53. $H = I^2 R t = 45000 \text{ J}$

54. $H = I^2 R t = 450 \text{ J}$

55. $H = I^2 R t = 14400 \text{ J}$

56. $H = I^2 R t = 16200 \text{ J}$

57. $H = I^2 R t = 5400 \text{ J}$

58. $H = I^2 R t = 12000 \text{ J}$

59. $H = I^2 R t = 75000 \text{ J}$

60. $H = I^2 R t = 600 \text{ J}$

61. $v \approx 16.15 \text{ cm}$ (using $1/v + 1/u = 1/f$ with sign convention)

62. $H = I^2 R t = 600 \text{ J}$

63. $H = I^2 R t = 8100 \text{ J}$

64. $H = I^2 R t = 21600 \text{ J}$

65. $H = I^2 R t = 1200 \text{ J}$

66. $H = I^2 R t = 4800 \text{ J}$

67. $H = I^2 R t = 40500 \text{ J}$

68. $H = I^2 R t = 11250 \text{ J}$

69. $H = I^2 R t = 1800 \text{ J}$

70. $H = I^2 R t = 5400 \text{ J}$

71. $H = I^2 R t = 37500 \text{ J}$

72. $H = I^2 R t = 1500 \text{ J}$

73. $H = I^2 R t = 24000 \text{ J}$

74. $H = I^2 R t = 1800 \text{ J}$

75. $H = I^2 R t = 15000 \text{ J}$

76. $H = I^2 R t = 19200 \text{ J}$

77. $H = I^2 R t = 24000 \text{ J}$

78. $H = I^2 R t = 3000 \text{ J}$

79. $H = I^2 R t = 4050 \text{ J}$

80. $H = I^2 R t = 4800 \text{ J}$

81. $H = I^2 R t = 2700 \text{ J}$

82. $H = I^2 R t = 2400 \text{ J}$

83. $H = I^2 R t = 9600 \text{ J}$

84. $H = I^2 R t = 7500 \text{ J}$

85. $H = I^2 R t = 1350 \text{ J}$

86. $H = I^2 R t = 3600 \text{ J}$

87. $H = I^2 R t = 1200 \text{ J}$

88. $H = I^2 R t = 600 \text{ J}$

89. $H = I^2 R t = 28800 \text{ J}$

90. $H = I^2 R t = 9600 \text{ J}$

91. $H = I^2 R t = 4800 \text{ J}$

92. $H = I^2 R t = 54000 \text{ J}$

93. $H = I^2 R t = 7500 \text{ J}$

94. $H = I^2 R t = 150 \text{ J}$

95. $H = I^2 R t = 19200 \text{ J}$

96. $H = I^2 R t = 2400 \text{ J}$

97. $H = I^2 R t = 30000 \text{ J}$

98. $H = I^2 R t = 96000 \text{ J}$

99. $H = I^2 R t = 5400 \text{ J}$

100. $H = I^2 R t = 27000 \text{ J}$